

INVERTING AND NON INVERTING AMPLIFIER

1. Inverting Amplifier

Aim:-

To design an inverting amplifier using an op-amp for the given gain of -2.

Apparatus Required:-

S.No	Components	Specification
1	Operational Amplifier	IC 741
2	Resistors	1 kΩ, 2 kΩ
3	Breadboard/Trainer Kit	1
4	CRO (Cathode Ray Oscilloscope)	1
5	Function Generator	1
6	Connecting Wires	As required

Design

For an inverting amplifier,

$$A_v = -\frac{R_f}{R_{in}}$$

Given,

$$A_v = -2$$

$$-2 = -\frac{R_f}{R_{in}}$$

Taking,

$$R_{in} = 1 \text{ k}\Omega$$

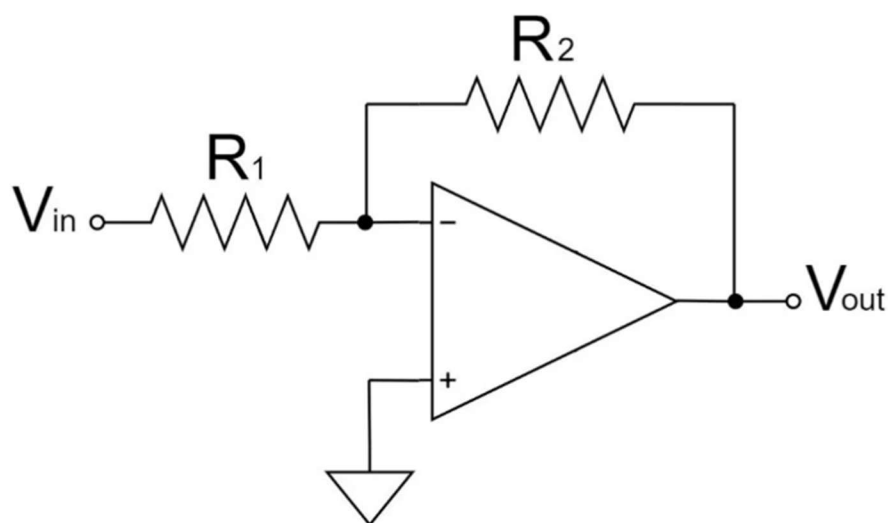
$$R_f = 2 \text{ k}\Omega$$

Output equation:

$$V_o = A_v V_{in}$$

$$V_o = -\frac{R_f}{R_{in}} V_{in}$$

Inverting Amplifier Circuit



Procedure

1. Connect IC 741 on the trainer kit.
2. Give dual power supply (+Vcc and -Vcc) to the IC.
3. Make the circuit connections as per the circuit diagram.
4. Apply sinusoidal input signal from the function generator to the inverting terminal.
5. Ground the non-inverting terminal.
6. Connect feedback resistor (R_f) between output and inverting terminal.
7. Switch ON the power supply and CRO.
8. Observe input and output waveforms on the CRO.
9. Verify that the output waveform is amplified and 180° out of phase with the input.
10. Tabulate the readings.

Observation Table:

S.No	V _{in} (V)	V _{out} (V)	Practical Gain	Theoretical Gain
1	1	-2.6	-2.6	-2.2
2	2	-5	-2.5	-2.2
3	3	-7.2	-2.4	-2.2
4	4	-9.4	-2.35	-2.2

Result

The working of the inverting amplifier was studied and verified successfully.

2. Non-Inverting Amplifier

Aim

To design a non-inverting amplifier using an op-amp for the given gain of 2.

Apparatus Required

S.No	Components	Specification
1	Operational Amplifier	IC 741
2	Resistors	1 kΩ, 2 kΩ
3	Function Generator	1
4	CRO	1
5	Dual Power Supply	±12 V
6	Trainer Kit	1
7	Connecting Wires	As required

Design

For a non-inverting amplifier,

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_{in}$$

Gain,

$$A_v = 1 + \frac{R_f}{R_1}$$

Given,

$$A_v = 2$$

$$2 = 1 + \frac{R_f}{R_1}$$

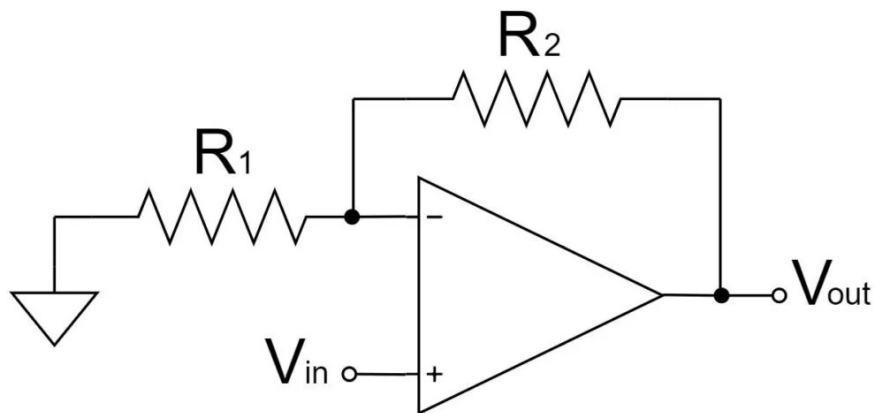
$$R_f = R_1$$

Taking,

$$R_1 = 1k\Omega$$

$$R_f = 1k\Omega$$

Non-Inverting Amplifier Circuit



Procedure

1. Place IC 741 on the trainer kit.
2. Connect dual power supply ± 12 V to the IC.
3. Make connections as per the circuit diagram.
4. Apply sinusoidal input signal from the function generator to the non-inverting terminal.
5. Connect feedback resistor (R_f) and resistor (R_1) properly.
6. Switch ON the power supply and CRO.
7. Observe input and output waveforms on the CRO.
8. Measure input voltage, output voltage, time period and frequency.
9. Verify that the output waveform is amplified and in phase with the input waveform.
10. Tabulate the readings.

Observation Table:

S.No	Vin (V)	Vout (V)	Practical Gain	Theoretical Gain
1	0.5	1.38	3.76	2.2
2	1	2.94	3.94	2.2
3	1.5	4.73	4.153	2.2
4	2	6.26	4.13	2.2
5	2.5	7.96	4.18	2.2
6	3	9.50	4.16	2.2
7	3.5	10.57	3.926	2.2

Result

The non-inverting amplifier was designed and verified successfully.